UIU CSE 4165 FINAL EXAM SURVIVAL GUIDE

Master Strategy, Exact Time Budget, Cheatsheet & Snippet Usage Guide

TOTAL TIME: 1h 55m
TOTAL MARKS: 30

1. Exam Time Management Plan (1 Hour 55 Minutes)

Students often fail to submit in time because they spend too much time designing or stuck on a syntax typo. Follow this strict minute-by-minute plan:

Time	Section	Marks	What to Do (Step-by-Step Action)
0:00 - 0:10 (10 mins)	**PRE-FLIGHT**	-	1. Open **XAMPP Control Panel** → Start **Apache** and **MySQL** (ensure green). 2. Create your exam working folder inside `C:\xampp\htdocs\web_final`. 3. Open VS Code in that folder. Test in browser: `http://localhost/web_final/`.
0:10 - 0:40 (30 mins)	**QUESTION 1** HTML + JavaScript	**10 M**	1. Create `index.html` (1-2 inputs, 1 button, 1 label below). (5 mins) 2. Create `script.js`: state variables outside function, `parseFloat()`, validation, running total, average, 4-tier ladder, warning limit. (20 mins) 3. Test **all sample input rows** given in question. (5 mins)
0:40 - 1:15 (35 mins)	**QUESTION 2** HTML Form + PHP	**10 M**	1. Create `form.html`: exactly 3 inputs, action `calculate.php`, method `POST`. (5 mins) 2. Write `calculate.php`: declare calculation function, `ceil()` for whole units, empty units, cost waste, 4-tier classification, target difference. (20 mins) 3. Print HTML `<table>` matching the sample output format. Test via `http://localhost/ ....` (10 mins)
1:15 - 1:45 (30 mins)	**QUESTION 3** SQL Setup + PHP & MySQL	**10 M**	1. **Write `database.sql` FIRST!** (50% marks): `CREATE DATABASE`, `CREATE TABLE`, and `INSERT` all seed rows. Import in phpMyAdmin. (10 mins) 2. Write `solution.php`: connect with `new mysqli()`, execute the 4 query tasks (GROUP BY, compound UPDATE, capped increment, advanced SELECT). (20 mins)
1:45 - 1:55 (10 mins)	**SUBMISSION**	-	1. Verify all 3 questions run without errors. 2. **Stop editing code!** Right-click `web_final` → Send to → Compressed (zipped) folder. 3. Upload `web_final.zip` to eLMS before deadline! (Zero marks for late/emailed code).

2. How to Use the A4 Double-Sided Cheat Sheet

Side 1: JavaScript (Q1) & PHP Logic (Q2)

- **Left Half (Q1)**: Contains complete HTML + JS skeleton. Has decoder table for running sum, averages, absolute difference, password regex, and array history.
- **Right Half (Q2)**: Contains complete HTML form + PHP calculation skeleton. Has decoder table for ceiling booking math, empty units, wasted money, and 4-tier status classification.
- **Workflow**: Read question → Find matching keyword in Decoder table → Copy skeleton → Replace variable names & numbers → Test with sample inputs.

Side 2: MySQL Setup (Q3) & Query Decoder

- **Left Half (Q3 Setup & PHP)**: Complete `database.sql` DDL/DML skeleton + complete `solution.php` script with connection, reusable table printer, and update runner.
- **Right Half (SQL Dictionary)**: Translates English exam phrases ("for each...", "only groups with...", "bonus with cap...", "10% discount", "compare to category average") directly into SQL!
- **Workflow**: Write `database.sql` → Import in phpMyAdmin → Write SQL query for each task → Paste into `solution.php`.

3. How to Use Keyboard Snippets in VS Code

All snippets are stored in `.vscode/web_exam.code-snippets` with **zero comments** (clean, pure code):

Type Prefix	Press	What It Inserts & How to Adapt
`examjs`	Tab	Inserts complete Q1 JavaScript tracker: change `GOAL`, threshold numbers, and feedback messages.
`examjsformula`	Tab	Inserts Q1 vitals/formula evaluator with arrays, `reduce` average, and `Math.abs`.
`examhtmlform`	Tab	Inserts clean 3-input HTML form with POST action to `calculate.php`.
`examphpcalc`	Tab	Inserts complete Q2 logistics engine: `ceil()`, empty capacity, cost waste, and efficiency tiers.
`examphptarget`	Tab	Inserts Q2 target performance evaluator: `$total - $target`, exact target strings, 4 tiers.
`examsqlsetup`	Tab	Inserts clean `CREATE DATABASE`, `CREATE TABLE` with all data types, and sample `INSERT`s.
`examphpmysql`	Tab	Inserts full Q3 PHP-MySQL script with table renderer, GROUP BY, capped UPDATE, and subquery CASE WHEN.

QUESTION-BY-QUESTION EXAM BLUEPRINT

Exact Structure, How to Write, and Traps to Avoid for Every Question

UIU EXAM PATTERNS

Question 1: HTML + JavaScript DOM & State Tracking (10 Marks)

The 6 Invariable Rules of Q1:

1. **HTML is minimal:** 1 or 2 `<input>` tags + 1 `<button>` + 1 `<label id="out">` placed directly below the button. No CSS needed!
2. **Memory must be outside function:** Variables that remember state (`let total = 0, count = 0, history = [];`) MUST be declared outside the click function. If you put them inside, they reset to 0 every click!
3. **Input Parsing:** Always use `parseFloat(document.getElementById("id").value)`. Check `if (isNaN(val) || val ≤ 0) return;`.
4. **4-Tier Classification Ladder:** Use `if ... else if ... else` checking upper or lower bounds. Check highest limit first if using ≥, or lowest limit first if using ≤.
5. **The Signature Warning Trap:** Every trimester has a condition like: "*If the user makes more than N attempts/entries without reaching the goal, display [Warning]!*"

```
if (count > 10 && total < GOAL) feedback += " Be cautious of frequent snacking!";
```

6. **Output to Label:** Use `document.getElementById("out").textContent = `...`;` or `.innerHTML`.

Question 2: HTML Form + PHP Business Logic (10 Marks)

The 5 Invariable Rules of Q2:

1. **Two files:** `form.html` (or `index.html`) and `calculate.php`. Every input MUST have a `name="..."` attribute (e.g. `name="val1"`).
2. **Always define a PHP function:** The question always says: "*Write a PHP function that takes three inputs...*". Do NOT write all code inline. Define `function process($a, $b, $c) { ... return [...]; }`.
3. **Ceiling Division for Whole Units:** When booking venues, movie screens, or carrying battery modules, partial units are forbidden:

```
$unitsNeeded = ceil($demand / $capacity);  // 150 / 60 = 2.5 → ceil() gives 3
```

4. **Empty Units & Wasted Money:**

```
$emptySeats = ($unitsNeeded * $capacity) - $demand;
$wastedMoney = $emptySeats * $pricePerSeat;  // Check sample output table for multiplier!
```

5. **Target Comparison Exact Strings:**

```
$d = $total - $target;
if ($d == 0)     $res = "Target met exactly (0 difference)";
elseif ($d > 0) $res = "Above target by $d";
else            $res = "Below target by " . abs($d);
```

Question 3: SQL Database Setup + PHP & MySQL Script (10 Marks)

The 5 Invariable Rules of Q3:

1. **Marks are split 50/50:** 5 Marks for `database.sql` and 5 Marks for `solution.php`. Hand in BOTH files!
2. **Write database.sql immediately:** Always start with `CREATE DATABASE IF NOT EXISTS`, `USE`, and `DROP TABLE IF EXISTS` so you can re-import freely.
3. **Standard Task 1 (Aggregation):** `SELECT category, COUNT(*) AS total FROM t GROUP BY category`. If filtered by count: `HAVING COUNT(*) > 1`. (NEVER put COUNT in WHERE!).
4. **Standard Task 2 & 3 (Updates with compound conditions & caps):**

```
-- Updating rating if salary < 40000 and not 'D'
UPDATE employee_final SET PerformanceRating = 'C' WHERE Salary < 40000 AND PerformanceRating ≠ 'D';

-- Bonus 5000 only if resulting salary ≤ 60000 (Test the NEW value in WHERE!)
UPDATE employee_final SET Salary = Salary + 5000 WHERE Salary > 50000 AND (Salary + 5000) ≤ 60000;
```

5. **Standard Task 4 (Subquery / Category Average comparison):**

```
SELECT name, category, price,
  CASE WHEN price > (SELECT AVG(price) FROM products t2 WHERE t2.category = t1.category)
       THEN 'Top Seller' ELSE 'Regular Seller' END AS seller_status
FROM products t1 ORDER BY price DESC;
```

Exam Day Emergency Fixes (When Something Breaks)

Symptom	Exact Cause & Instant Fix
JS button click does nothing	Open DevTools (F12 → Console). Look for red error: typo in `getElementById` or function name mismatch.
Output shows NaN	Empty input box or forgot `parseFloat()` on the value string.
Raw PHP code displayed on screen	You double-clicked the `.php` file. Open via `http://localhost/web_final/` ... in your browser!
Blank white page in PHP	PHP fatal syntax error. You missed a semicolon `;` or closing brace `}` or quote `'`. Check last line edited.
MySQL table already exists / duplicate key	Re-import `database.sql` in phpMyAdmin (the script contains `DROP TABLE IF EXISTS`).